



NQRUST-K8s

Enterprise Kubernetes | Indonesian Support

Enterprise-grade Kubernetes platform engineered in Indonesia. Built on globally proven foundations, designed for national-scale data management.

64+
WORKER
NODES

Multi
CLUSTER
MGMT

CIS
HARDENED
K8S

24/7
LOCAL
SUPPORT

Version 1.0 – Brochure
May 2026

Introduction

What Is NQRust-K8s?

An enterprise-grade Kubernetes platform with multi-cluster management, CIS security baseline, and local Indonesian technical support.

NQRust-K8s is an enterprise container platform built on production-proven, Kubernetes-native foundations. It is designed for organizations running **national-scale digital services** that demand reliability, security, and operational control. Based on standard Kubernetes, the platform eliminates dependency on platform-specific capabilities that increase vendor lock-in.

The platform provides **multi-cluster management** from a single centralized console, enterprise security baselines meeting CIS Kubernetes Benchmark, and a complete ecosystem spanning container registry, CI/CD pipelines, monitoring, logging, and backup. With a **local Indonesian technical team**, SLA in Bahasa Indonesia, and response times aligned to local timezone, NQRust-K8s delivers the responsive support that global vendor mechanisms cannot match.

Core Pillars



Enterprise-Grade

Production Kubernetes with enterprise security baseline for hundreds to thousands of containers. Embedded etcd for HA without external datastore. Containerd runtime with no Docker dependency. CIS Benchmark compliance with minimal manual configuration.



Multi-Cluster Management

Manage all clusters from a single centralized console. Automated cluster provisioning. Multi-datacenter ready. Import existing clusters from any provider. Unified dashboard for visibility across all environments from dev to production.



Local Indonesian Support

Indonesian technical team with SLA in Bahasa Indonesia. Response aligned to WIB/WITA/WIT timezone. 8x5 or 24x7 support options per operational needs. On-site knowledge transfer for operations teams. No dependency on global vendor support mechanisms.

Complete Platform Stack

Every layer built on Kubernetes-native components

Layer	Components
Console	NQRust Dashboard, Monitoring & Alerts, Multi-Cluster Management
Platforms	Harbor Registry, Argo CD / Tekton CI/CD, Keycloak SSO, Velero Backup, App Catalog (Helm)
Observability	Prometheus, Grafana, Loki, Fluent Bit, Alertmanager
Security	Kyverno / OPA Policy Engine, cert-manager, NGINX Ingress, NetworkPolicy, External Secrets
Kubernetes	NQRust-K8s Engine (Enterprise Kubernetes), Longhorn / Rook-Ceph Storage, Calico / Cilium CNI
Infrastructure	Bare metal or VM — compatible with existing Dell, HP, Lenovo hardware. NAS/SAN, 100Gbps networking.

Deployment & Day-2 Operations



Deployment

Deploy on bare metal, VMs, or hybrid environments. Automated cluster provisioning with HA control plane (3+ masters). Compatible with existing Dell, HP, and Lenovo hardware. Air-gapped deployment supported for classified environments. CSI drivers for Dell PowerFlex, NAS (NFS/S3), SAN, Longhorn, and Rook-Ceph.



Day-2 Operations

Zero-downtime rolling upgrades keep services available during maintenance. Patches and security fixes delivered continuously. Centralized monitoring (Prometheus + Grafana), logging (Loki + Fluent Bit), and alerting (Alertmanager) out-of-the-box. Velero backup per namespace. 8x5 or 24x7 local Indonesian support with on-site knowledge transfer.

Platform Capabilities

Features That Power National Operations

Enterprise-grade capabilities for national-scale digital services.



Security & Identity

Pod Security Admission + Kyverno policy engine. Enterprise-grade RBAC. Keycloak SSO with OIDC, LDAP, and SAML. Automated certificate management via cert-manager.



Monitoring & Logging

Real-time dashboards with Prometheus + Grafana. Centralized logging via Loki + Fluent Bit. Alertmanager for automated notifications to operations teams.



Backup & DR

Velero for per-namespace Kubernetes-layer backup. Existing backup solutions can be integrated. Restore validated at every deployment phase.



Provisioning & Scaling

Automated cluster provisioning. HPA, VPA, and Cluster Autoscaler for hundreds to thousands of containers. Ready to scale from 32 to 64+ worker nodes.



Container Registry

Harbor enterprise registry with vulnerability scanning and multi-site replication. Webhook integration for automated image deployment pipelines.



Developer Experience

App Catalog (Helm), GitOps pipeline (Argo CD), YAML editor, exec terminal. Familiar workflow for development teams working with ReactJS, Java, and Node.js.



Ingress & Networking

NGINX Ingress controller. Calico / Cilium CNI for network policy enforcement. NetworkPolicy for microsegmentation. Load balancing for national-scale traffic.



Multi-Cluster

Manage all clusters from a single console. Built-in at no additional cost. Multi-datacenter ready. Unified visibility across all environments.



Storage Compatibility

CSI drivers compatible with Dell PowerFlex, NAS (NFS/S3), SAN, and existing enterprise storage. Longhorn and Rook-Ceph as distributed storage options.



Auto-Scaling

HPA + VPA + Cluster Autoscaler. Resources automatically adjust to national traffic loads without manual intervention. Zero-downtime rolling upgrades.



High Availability

HA control plane with 3+ masters and embedded etcd. No single point of failure. Rolling upgrades without downtime. Containerd runtime.



Workload Compatibility

ReactJS, Java/Spring Boot, Node.js workloads run without code changes. Oracle Database stays on bare metal. Automatic manifest conversion tooling provided.



CIS Benchmark & Compliance

Built on an enterprise Kubernetes distribution that **passes CIS Kubernetes Benchmark with minimal operator intervention**. Containerd runtime with no Docker dependency. Embedded etcd eliminates external datastore complexity. FIPS-capable cryptographic modules. Hardened container images scanned for vulnerabilities. Security policies enforced via Kyverno / OPA across all namespaces.



Scalability & Performance

From **32 servers today to 64+ servers tomorrow**. 100Gbps inter-node switching. HPA + VPA + Cluster Autoscaler for automatic resource adjustment under national-scale traffic loads. HA control plane with 3+ masters ensures no single point of failure. Zero-downtime rolling upgrades keep services available during maintenance.

Platform Comparison

Aspect	Proprietary Platforms	NQRust-K8s
Cost	Significant annual subscription	More efficient, transparent pricing
Lock-in	Platform-specific features	Kubernetes-native, open standards
Support	Global vendor mechanisms	Indonesian team, local SLA, fast response
Multi-Cluster	Separate additional cost	Built-in, no extra charge
Customization	Constrained by platform	Flexible, tailored to requirements

Use Cases

Built for National-Scale Operations



National Digital Services

Container platform for services accessed across **borders, ports, and airports nationwide**. Hundreds to thousands of containers with auto-scaling. HA control plane for 24/7 services with no single point of failure.



Enterprise Container Platform

Replace proprietary container platforms with **Kubernetes-native standards**. Complete ecosystem from registry to monitoring to backup. Predictable licensing without per-socket or per-core pricing complexity.



Multi-Datacenter & Scaling

From **32 servers today to 64+ tomorrow**. Multi-cluster management from one console. 100Gbps inter-node switching. CSI drivers compatible with Dell PowerFlex and existing NAS/SAN infrastructure.



Security & Compliance

CIS Kubernetes Benchmark compliance with minimal configuration. Pod Security Admission + Kyverno. Enterprise RBAC. Keycloak SSO. Vulnerability scanning via Harbor. Audit-ready platform.

Under The Hood

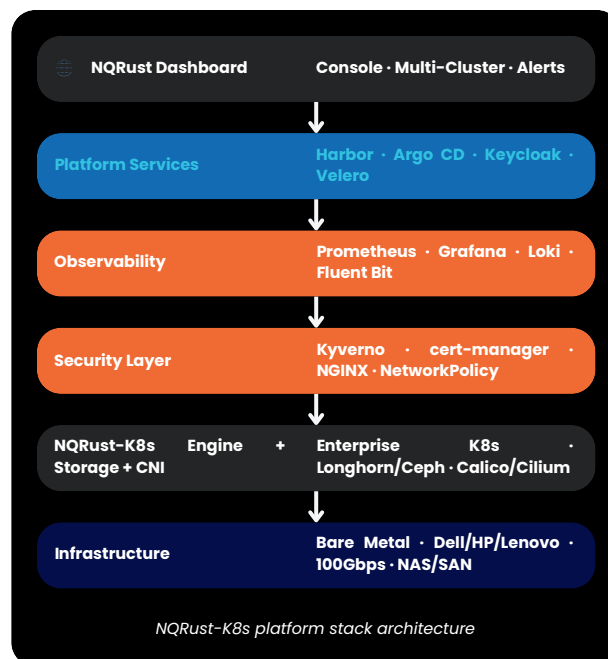
Enterprise-grade Kubernetes-native components proven in global production environments.

The **NQRust-K8s Engine** serves as the platform foundation — an enterprise Kubernetes distribution focused on security and compliance with embedded etcd, containerd runtime, and CIS Benchmark defaults. No Docker dependency, closely aligned with upstream Kubernetes.

The **observability layer** uses Prometheus for metrics, Grafana for dashboards, Loki for log aggregation, Fluent Bit for collection, and Alertmanager for automated notifications. All integrated out-of-the-box.

The **security layer** includes Pod Security Admission, Kyverno policy engine, cert-manager, NGINX Ingress, NetworkPolicy, and External Secrets. **Keycloak** provides SSO with OIDC, LDAP, and SAML support.

Velero handles per-namespace backup at the Kubernetes layer. **Harbor** provides enterprise container registry with vulnerability scanning. **Argo CD** enables GitOps pipeline for declarative, version-controlled deployments.



Why NQRust-K8s

Enterprise Kubernetes engineered in Indonesia — reliable, scalable, and cost-efficient.



Cost Efficiency

More efficient and transparent pricing compared to proprietary platform subscriptions. Multi-cluster management built-in at no additional cost. Predictable mid-term budget planning.



Zero Vendor Lock-in

Built on Kubernetes-native standards. No dependency on a single vendor ecosystem. Flexible platform control and customization tailored to your requirements, not platform limitations.



Local Indonesian Support

Indonesian technical team. SLA in Bahasa Indonesia. Response aligned to local timezone. On-site knowledge transfer. 8x5 or 24x7 support per contract.



Production-Proven Stack

Every component in the stack is production-proven at global scale. CIS Benchmark compliance. Embedded etcd for HA. Containerd runtime. Hardened images. Enterprise-grade from day one.